Circles and Parabolas

ANSWER KEY

Section 1: Standard Equation and Features of Circles

1 Find the center and radius of the circle $(x-3)^2+(y+5)^2=49$.

a. **Center:** $(3,\ -5)$**, Radius:** 7 ✓

b. Center: $(-3, 5)$, Radius: 7

c. Center: $(3,\ -5)$, Radius: 49

d. Center: $(-3, 5)$, Radius: 49

Standard form is $(x-h)^2+(y-k)^2=r^2$. Here $h=3, k=-5, r=\sqrt{49}=7$.

2 Convert the general equation $x^2+y^2-6x+8y-11=0$ to standard form by completing the square.

a. $(x-3)^2+(y+4)^2=36$ ✓

b. $(x+3)^2+(y-4)^2=36$

c. $(x-3)^2+(y+4)^2=25$

d. $(x-6)^2+(y+8)^2=11$

$(x^2-6x+9)+(y^2+8y+16)=11+9+16=36 \Rightarrow (x-3)^2+(y+4)^2=36$.

3 Find the equation of the circle passing through the origin with center $(-2, 3)$.

a. $(x+2)^2+(y-3)^2=13$ ✓

b. $(x-2)^2+(y+3)^2=13$

c. $(x+2)^2+(y-3)^2=5$

d. $(x+2)^2+(y-3)^2=25$

Radius $r=\sqrt{(-2-0)^2+(3-0)^2}=\sqrt{4+9}=\sqrt{13}$. Equation: $(x+2)^2+(y-3)^2=13$.

Section 2: Parabolas with Vertical Axis of Symmetry

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4 Find the focus and directrix of the parabola $x^2 = 12y$.

a. **Focus:** $(0, 3)$**, Directrix:** $y = -3$ ✓

b. Focus: $(3, 0)$, Directrix: $x = -3$

c. Focus: $(0, 12)$, Directrix: $y = -12$

d. Focus: $(0, 6)$, Directrix: $y = -6$

$x^2 = 4py \Rightarrow 4p = 12 \Rightarrow p = 3$. Vertical axis, opens up: Focus is $(0, p) = (0, 3)$, directrix is $y = -p = -3$.

5 Find the standard equation of the parabola with vertex $(0, 0)$ and focus $(0, -2)$.

a. $x^2 = -8y$ ✓

b. $y^2 = -8x$

c. $x^2 = 8y$

d. $y^2 = -4x$

Opens downward: $p = -2$. Standard equation: $x^2 = 4py = 4(-2)y = -8y$.

6 Find the focal diameter (latus rectum length) of $(x - 1)^2 = -16(y + 2)$.

a. 16 ✓

b. 4

c. 8

d. 32

The focal diameter is $|4p| = |-16| = 16$.

Section 3: Parabolas with Horizontal Axis of Symmetry

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7 Find the vertex, focus, and directrix of $(y-2)^2 = 8(x+1)$.

a. **Vertex:** $(-1, 2)$**, Focus:** $(1, 2)$**, Directrix:** $x = -3$ ✓

b. Vertex: $(1, -2)$, Focus: $(3, -2)$, Directrix: $x = -1$

c. Vertex: $(-1, 2)$, Focus: $(-1, 4)$, Directrix: $y = 0$

d. Vertex: $(2, -1)$, Focus: $(4, -1)$, Directrix: $x = 0$

$4p = 8 \Rightarrow p = 2$. Vertex is $(h, k) = (-1, 2)$. Opens right: Focus is $(h + p, k) = (1, 2)$, directrix is $x = h - p = -1 - 2 = -3$.

8 Find the equation of a parabola with vertex $(3, -1)$ and focus $(1, -1)$.

a. $(y+1)^2 = -8(x-3)$ ✓

b. $(x-3)^2 = -8(y+1)$

c. $(y+1)^2 = 8(x-3)$

d. $(y-3)^2 = -4(x+1)$

Horizontal axis $y = -1$. Opens left because focus $x = 1$ is to the left of vertex $x = 3$. $p = 1 - 3 = -2$. $(y+1)^2 = 4(-2)(x-3) = -8(x-3)$.

9 Convert $y^2 + 4y - 4x + 8 = 0$ to standard form.

a. $(y+2)^2 = 4(x-1)$ ✓

b. $(y-2)^2 = 4(x+1)$

c. $(y+2)^2 = 4x - 8$

d. $(x-1)^2 = 4(y+2)$

$(y^2 + 4y + 4) = 4x - 8 + 4 = 4x - 4 \Rightarrow (y+2)^2 = 4(x-1)$.

Section 4: Reflective Properties and Applications

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10 A satellite dish has the shape of a paraboloid. If the cross-section is $x^2 = 20y$ (in inches), where should the receiver be placed?

a. **At $(0, 5)$, which is 5 inches above the vertex ✓**

b. 20 inches above the vertex

c. 10 inches above the vertex

d. 2.5 inches above the vertex

Signals parallel to the axis reflect to the focus. $4p = 20 \Rightarrow p = 5$ inches above the vertex.

11 A suspension bridge cable hangs in the shape of a parabola with towers 100 m apart and 20 m high, touching the road at the center. Find its equation with origin at the vertex.

a. $y = \frac{1}{125}x^2$ ✓

b. $y = \frac{1}{250}x^2$

c. $y = \frac{1}{50}x^2$

d. $y = \frac{1}{20}x^2$

Vertex at $(0, 0)$. Towers at $(\pm 50, 20)$. $20 = a(50)^2 = 2500a \Rightarrow a = \frac{20}{2500} = \frac{1}{125}$.

12 Why do parabolic headlights use a light source placed exactly at the focus?

a. **Light rays reflecting off the parabolic surface emerge in a parallel beam ✓**

b. It spreads the light in all directions equally

c. It minimizes heat production

d. It prevents the mirror from tarnishing

By the geometric reflective property of parabolas, all rays emanating from the focus reflect parallel to the axis of symmetry.

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